MODEL LAB · 2026-10-23

Every prior, same slate

The held-out scores that decided which prior is served, then how each one would have ranked today's hitters.

Served (full season)
served prior

11
priors compared

23,148
held-out plate appearances

3.03%
held-out HR rate

0.6081
best AUC · KNN comps

Held-out scores

Fitted on plate appearances before 2026-09-07, scored on the 23,148 that came after (actual HR rate 3.03%). No outcome in the test window was visible during fitting. **Read AUC first** — the slate ranks hitters and takes the top six, so ordering is what matters. Brier and log loss are dominated by the ~3% base rate and barely separate the priors. Metrics are per plate appearance; the probabilities lower down are per game.

PRIOR	AUC	TOP-DECILE LIFT	BRIER	LOG LOSS	CAL. RMSE
Flat league League HR rate as the prior for every hitter.	0.5917	1.67×	0.02934	0.13484	0.00602
KNN comps Prior is the pooled rate of the nearest comparables.	0.6081	1.52×	0.02929	0.13367	0.00460
SVR Support-vector regression on the same feature matrix.	0.6033	1.57×	0.02931	0.13421	0.00492
Random forest Forest regression; a learned similarity metric.	0.6055	1.55×	0.02929	0.13369	0.00375
Core (KNN+SVR) Blend of KNN and SVR, no forest.	0.6081	1.52×	0.02929	0.13367	0.00460
Forest blend KNN + SVR + forest, weights fitted out-of-fold.	0.6081	1.52×	0.02929	0.13367	0.00460
Linear regression Ordinary least squares on the same features.	0.6059	1.57×	0.02930	0.13385	0.00369
Logistic regression Binomial logistic on the per-PA home run outcome.	0.6053	1.57×	0.02932	0.13396	0.00541
Neural net (torch) Two-layer PyTorch MLP, PA-weighted, trained on the same features.	0.6056	1.62×	0.02930	0.13381	0.00385
XGBoost Gradient-boosted trees, depth 3 with subsampling and L2.	0.6036	1.55×	0.02930	0.13393	0.00411

BLEND WEIGHTS (FOREST VARIANT)

KNN **1.00** SVR **0.00** RF **0.00**

SHRINKAGE K PER PRIOR

league **259** knn **259** svr **259** rf **259** core **259** ensemble **259** linear **259** logistic **259** nn **259** xgb **259**

Estimated by method of moments on out-of-fold residuals, not hand-tuned.

Thin-sample subset

The same held-out window, restricted to hitters with fewer than 100 plate appearances before the cutoff (2,370 PA, actual HR rate 2.49%). For a regular with 600 PA the observed record swamps any prior, so the priors mostly separate here.

PRIOR	AUC	TOP-DECILE LIFT	BRIER	LOG LOSS	CAL. RMSE
Flat league League HR rate as the prior for every hitter.	0.5763	1.19×	0.02428	0.11650	0.01359
KNN comps Prior is the pooled rate of the nearest comparables.	0.6166	2.03×	0.02418	0.11439	0.01018
SVR Support-vector regression on the same feature matrix.	0.5967	1.36×	0.02433	0.11860	0.01890
Random forest Forest regression; a learned similarity metric.	0.6122	1.69×	0.02423	0.11481	0.01426
Core (KNN+SVR) Blend of KNN and SVR, no forest.	0.6166	2.03×	0.02418	0.11439	0.01018
Forest blend KNN + SVR + forest, weights fitted out-of-fold.	0.6166	2.03×	0.02418	0.11439	0.01018
Linear regression Ordinary least squares on the same features.	0.6137	1.69×	0.02425	0.11594	0.01351
Logistic regression Binomial logistic on the per-PA home run outcome.	0.6211	1.86×	0.02429	0.11568	0.01283
Neural net (torch) Two-layer PyTorch MLP, PA-weighted, trained on the same features.	0.6118	2.03×	0.02421	0.11509	0.01040
XGBoost Gradient-boosted trees, depth 3 with subsampling and L2.	0.5986	1.53×	0.02431	0.11614	0.01566

Slate agreement

PRIOR	SHARED	ITS TOP SIX
Served (full season)	6/6	Jackson Chourio 16.1%, Jake Bauers 16.1%, William Contreras 14.4%, Junior Caminero 13.8%, Christian Yelich 13.3%, Jonathan Aranda 13.1%
Flat league	6/6	Jackson Chourio 16.3%, Jake Bauers 15.1%, William Contreras 14.3%, Christian Yelich 13.5%, Junior Caminero 13.2%, Jonathan Aranda 12.5%
KNN comps	6/6	Jake Bauers 16.3%, Jackson Chourio 16.3%, William Contreras 14.4%, Christian Yelich 13.7%, Junior Caminero 13.7%, Jonathan Aranda 13.2%
SVR	6/6	Jackson Chourio 16.9%, Jake Bauers 16.5%, Junior Caminero 14.9%, William Contreras 14.4%, Christian Yelich 13.1%, Jonathan Aranda 13.0%
Random forest	6/6	Jackson Chourio 16.8%, Jake Bauers 16.7%, William Contreras 14.5%, Junior Caminero 14.1%, Christian Yelich 13.6%, Jonathan Aranda 12.9%
Core (KNN+SVR)	6/6	Jake Bauers 16.3%, Jackson Chourio 16.3%, William Contreras 14.4%, Christian Yelich 13.7%, Junior Caminero 13.7%, Jonathan Aranda 13.2%
Forest blend	6/6	Jake Bauers 16.3%, Jackson Chourio 16.3%, William Contreras 14.4%, Christian Yelich 13.7%, Junior Caminero 13.7%, Jonathan Aranda 13.2%
Linear regression	6/6	Jackson Chourio 16.8%, Jake Bauers 16.2%, William Contreras 14.4%, Junior Caminero 13.9%, Christian Yelich 13.7%, Jonathan Aranda 13.1%
Logistic regression	6/6	Jackson Chourio 16.9%, Jake Bauers 16.2%, William Contreras 14.4%, Junior Caminero 14.2%, Christian Yelich 13.2%, Jonathan Aranda 12.9%
Neural net (torch)	6/6	Jackson Chourio 16.9%, Jake Bauers 16.4%, William Contreras 14.4%, Junior Caminero 14.2%, Christian Yelich 13.4%, Jonathan Aranda 13.0%
XGBoost	6/6	Jackson Chourio 17.0%, Jake Bauers 16.8%, William Contreras 14.7%, Junior Caminero 14.5%, Christian Yelich 13.2%, Jonathan Aranda 13.1%

Hitter by hitter

The top 40 bats by the served probability.

	HITTER		SERVED (FULL SEASON)	FLAT LEAGUE	KNN COMPS	SVR	RANDOM FOREST	CORE (KNN+SVR)	FOREST BLEND	LINEAR REGRESSION	LOGISTIC REGRESSION	NEURAL NET (TO
1	**Jackson Chourio**	MIL	**16.1%**	16.3%	16.3%	16.9%	16.8%	16.3%	16.3%	16.8%	16.9%	
2	**Jake Bauers**	MIL	**16.1%**	15.1%	16.3%	16.5%	16.7%	16.3%	16.3%	16.2%	16.2%	
3	**William Contreras**	MIL	**14.4%**	14.3%	14.4%	14.4%	14.5%	14.4%	14.4%	14.4%	14.4%	
4	**Junior Caminero**	TB	**13.8%**	13.2%	13.7%	14.9%	14.1%	13.7%	13.7%	13.9%	14.2%	
5	**Christian Yelich**	MIL	**13.3%**	13.5%	13.7%	13.1%	13.6%	13.7%	13.7%	13.7%	13.2%	
6	**Jonathan Aranda**	TB	**13.1%**	12.5%	13.2%	13.0%	12.9%	13.2%	13.2%	13.1%	12.9%	
7	**Garrett Mitchell**	MIL	**12.9%**	12.4%	12.6%	11.5%	12.3%	12.6%	12.6%	12.4%	12.2%	
8	**Brice Turang**	MIL	**12.1%**	12.3%	12.0%	11.9%	11.9%	12.0%	12.0%	11.5%	11.4%	
9	**Yandy Díaz**	TB	**11.1%**	11.4%	11.1%	11.5%	11.4%	11.1%	11.1%	11.3%	11.1%	
10	**Victor Mesa Jr.**	TB	**10.1%**	10.0%	10.1%	10.6%	10.5%	10.1%	10.1%	10.4%	10.4%	
11	**David Hamilton**	MIL	**7.1%**	8.7%	6.9%	6.7%	6.4%	6.9%	6.9%	7.1%	6.9%	
12	**Jonny DeLuca**	TB	**6.6%**	7.2%	6.5%	6.5%	6.0%	6.5%	6.5%	6.5%	6.6%	
13	**Sal Frelick**	MIL	**6.6%**	8.3%	6.7%	6.2%	6.2%	6.7%	6.7%	6.4%	6.5%	
14	**Liam Hicks**	TB	**6.5%**	7.6%	6.5%	7.3%	6.7%	6.5%	6.5%	6.9%	6.8%	
15	**Cooper Pratt**	MIL	**6.3%**	8.2%	6.4%	5.5%	5.9%	6.4%	6.4%	5.5%	5.7%	
16	**Ben Williamson**	TB	**4.0%**	6.6%	4.1%	4.0%	3.9%	4.1%	4.1%	2.9%	4.2%	
17	**Taylor Walls**	TB	**4.0%**	5.5%	4.0%	3.2%	3.2%	4.0%	4.0%	4.1%	4.1%	
18	**Chandler Simpson**	TB	**2.7%**	5.9%	3.4%	1.5%	1.6%	3.4%	3.4%	1.5%	3.1%	

Per-game probabilities: each prior's per-PA rate run through this hitter's own plate appearances, park, weather, bullpen and times-through-the-order terms. Only the prior changes across the columns — every other term is held fixed, so a left-to-right spread is the prior disagreeing about the hitter, not the context. Shading grows with the probability.

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